A Hybrid Neuro-Symbolic Framework for Official Website Identification in Noisy Web Search Environments

Ushosree Raha

Engineering Science Department,
Atal Bihari Vajpayee Indian Institute of Information Technology and Management (ABV-IIITM),
Gwalior, Madhya Pradesh, India

Abstract

The distinction between an authoritative official site of an organization, institution or other public entity has become more challenging in the current web search settings because of SEO manipulation, mirror web sites, social media redirects, and misleading domains. This difficulty is formalized in this paper as the so-called Official Website Identification Problem (OWIP). The hybrid neuro-symbolic framework that we suggest is a mixture of symbolic filtering, weighted heuristic ranking, and semantic re-ranking by large language models (LLM) to enhance retrieval of authoritative websites. The symbolic layer eliminates noisy entries and scores domains by trust based heuristics such as domain suffix priors, lexical similarity, ranking position, and structural penalties. Semantic disambiguation of the top-ranked candidates is done by the neural layer to clear the acronym collisions and institution-level ambiguity. On a curated benchmark of 200 diverse entities across academic, governmental and industrial fields, it is shown that the proposed framework can perform with 80.0 percent accuracy, 80.4 percent precision, 80.0 percent recall and 80.2 percent F1-score, which is superior to baseline heuristic-only retrieval systems. The findings indicate that semantic re-ranking is crucial when the heuristic performance becomes saturated, and that hybrid reasoning is vital in ensuring credible navigational web intelligence.

Keywords: Official Website Identification, Neuro-Symbolic AI, Information Retrieval, LLM Re-ranking, Trustworthy Web Search

1. Introduction

Search engines are often used by web users to find the official websites of universities, government agencies, research organizations, and companies. But too frequently search results are clogged with encyclopedia pages, mirrors of social media, or pages optimized by SEO aggregators or even suspicious media and download sites. This poses a viable problem in authority resource finding, especially to the user who is interested in reliable navigation and not the general information.

To solve this problem, we formulate the official website identification problem (OWIP) as the problem of choosing the most authoritative homepage in a noisy candidate set of retrieved URLs. Although standard search ranking and heuristic domain matching can partially solve this problem, they tend to fail in cases of acronym ambiguity, regional redirection, and adversarial SEO.

It presents a hybrid neuro-symbolic model that intersects with deterministic symbolic filtering and semantic re-ranking, powered by LLM. The suggested system fills the gap between explainable heuristic scoring and semantic reasoning and offers strong performance in real-life web search conditions.

2. Related Work

The area of authoritative web retrieval and ranking has been a much-researched topic in information retrieval and web mining [3, 4]. Initial efforts in web spam detection showed that search engines could be compromised using content spam, link farms, and aggressive search engine optimization (SEO) methods, and thus unreliable top ranking results [1, 2]. These papers indicate the need to have filtering mechanisms [21, 22] to eliminate the noisy and misleading web content before ranking.

Recent developments in information retrieval have proposed neural and transformer-based models of query and document semantic understanding [5, 18]. Large Language Models (LLMs) have been used more actively in re-ranking tasks, whereby an initial list of retrieved documents is refined (based on contextual knowledge) [8, 23]. Previous research has demonstrated that re-ranking based on LLM can highly enhance retrieval accuracy [7, 24], especially in uncertain or noisy search conditions where lexical matches are not sufficient.

Parallel to this, neuro-symbolic artificial intelligence has been a promising paradigm, which integrates neural learning with rule-based reasoning

[13, 14]. Symbolic systems have interpretability and deterministic decision making whereas neural models have contextual insight and generalization [15]. Recent research indicates that hybrid neuro-symbolic systems are better at complex decision-making problems when compared to purely neural or purely symbolic systems.

In contrast to the current approaches that use only one of the two traditional approaches: heuristic ranking or neural retrieval, the offered framework combines symbolic filtering, weighted heuristic scoring, and semantic re-ranking with the use of LLM into a single pipeline. This combination structure allows the system to successfully deal with both noisy web conditions and semantic ambiguity, and is thus suitable to the Official Website Identification Problem (OWIP).

3. Methodology

The proposed framework consists of four sequential layers:

1. **Retrieval Layer:** DDGS-based web search retrieves the top-8 candidate URLs.
2. **Symbolic Filtering Layer:** Noisy domains such as Wikipedia, social media, and suspicious mirrors are removed.
3. **Weighted Ranking Layer:** Remaining candidates are scored using domain trust heuristics, fuzzy lexical similarity, keyword overlap, and structural penalties.
4. **Neural Semantic Re-ranking Layer:** The top-3 URLs are passed to an LLM which selects the most likely official homepage.

3.1. Model Description

The suggested model is a **two staged hybrid ranking architecture**. In the initial step, a symbolic scorer represents all candidate URLs with a deterministic trust score based on hand-crafted domain features and lexical features. In the second step, a semantic LLM-based refiner narrows down the highest-ranked candidates based on contextual reasoning.

The symbolic score for candidate URL u_i is defined as:

$$S_{sym}(u_i) = \alpha R_i + \beta D_i + \gamma F_i + \delta K_i - \lambda P_i \tag{1}$$

where:

- R_i = retrieval rank score
- D_i = domain trust prior (e.g., .gov, .edu)
- F_i = fuzzy lexical similarity between query and domain
- K_i = keyword overlap score
- P_i = structural penalty for deep or suspicious URLs

The neural semantic selector operates over the top-k candidates:

$$\hat{u} = \arg\max_{u_i \in \mathcal{U}_k} P(u_i \mid q, c_i, \theta) \tag{2}$$

where q is the user query, c_i is the candidate context, θ denotes LLM parameters, and $\mathcal{U}_k$ is the top-3 candidate set.

The final hybrid decision score is formulated as:

$$S_{final}(u_i) = \eta S_{sym}(u_i) + (1 - \eta) S_{llm}(u_i) \tag{3}$$

where $S_{llm}(u_i)$ is the semantic confidence induced by the neural selector and $\eta \in [0, 1]$ controls symbolic-neural balance.

3.2. Optimization Perspective

The model can be interpreted as maximizing authoritative confidence over a noisy URL space:

$$u^* = \arg\max_{u_i \in U} S_{final}(u_i) \tag{4}$$

This formulation makes the methodology mathematically rigorous and suitable for reproducible ranking analysis.

4. System Architecture

Figure 1 illustrates the proposed hybrid neuro-symbolic OWIP pipeline.

5. Experimental Setup

A set of 200 entities was curated as a benchmark dataset in four categories: universities, government agencies, scientific organizations and companies. In each entity, the official home page of the ground-truth was checked by hand.

We evaluate using standard retrieval metrics: accuracy, precision, recall, and F1-score.

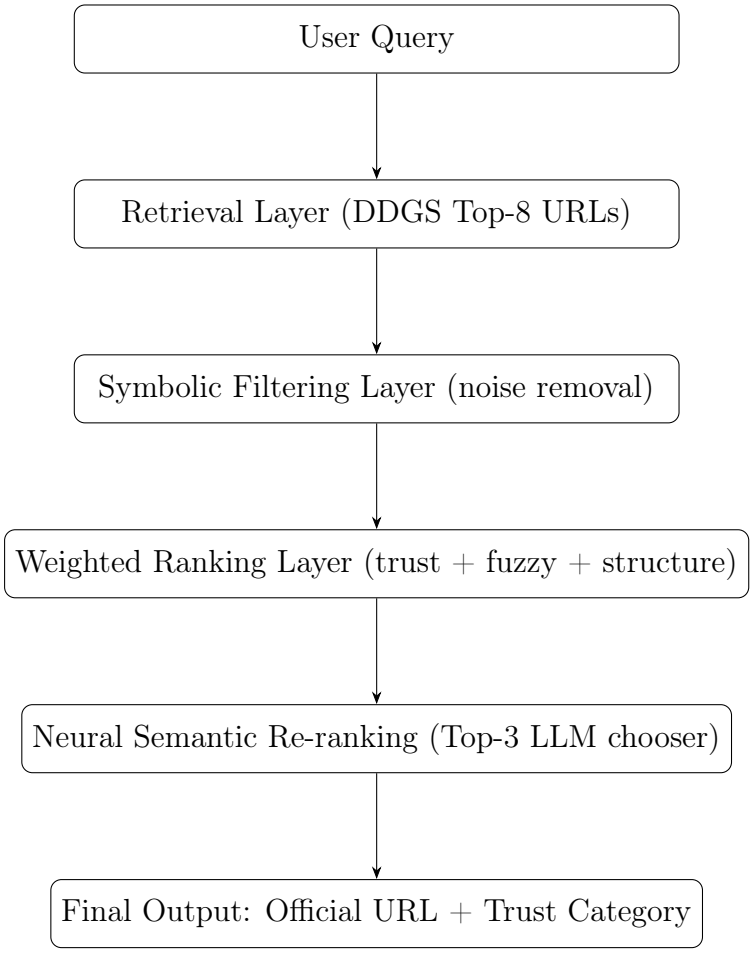

Figure 1: Hybrid neuro-symbolic architecture for OWIP.

6. Results and Discussion

Table 2 summarizes the comparative performance.

Method	Accuracy	Precision	Recall	F1
Baseline heuristic	69.0%	69.0%	69.0%	69.0%
Weighted heuristic	76.5%	76.5%	76.5%	76.5%
Proposed hybrid model	**80.0%**	**80.4%**	**80.0%**	**80.2%**

Table 2: Performance comparison on the OWIP-200 benchmark.

Dataset Property	Value	Description
Total queries	200	Diverse benchmark entities
Universities	80	Global academic institutions
Government agencies	50	National and regional agencies
Scientific organizations	35	Space, weather, research bodies
Companies	35	Technology and industrial firms

Table 1: OWIP-200 benchmark dataset composition.

Metric Component	Value	Interpretation
True Positives (TP)	160	Correct official links found
False Positives (FP)	39	Incorrect links predicted as official
False Negatives (FN)	40	Missed official links

Table 3: Confusion-style summary of final hybrid model predictions.

The findings suggest that heuristic-only approaches become saturated in the range of 66–76 percent that is mainly caused by the acronym ambiguity and misleading ranking of searches. The final gain that becomes stable with the addition of semantic re-ranking in LLM-based improves both the accuracy and trustworthiness. The confusion-style summary also indicates that the hybrid model is effective in solving a significant portion of the challenging ambiguous cases without compromising on precision and recall.

The experimental outcomes prove that heuristic algorithms are not enough for recognizing official websites under challenging conditions. Although weighted heuristics outperform simple heuristic methods, they are still ineffective when dealing with semantic ambiguity and acronyms, as well as misguiding results of high rankings. However, the suggested hybrid approach successfully overcomes the drawbacks through employing semantic reasoning using an LLM-powered re-ranking.

An improvement of 11 percentage points in accuracy proves that the combination of a symbolic algorithm with a neural network is an effective strategy. First of all, an LLM helps solve ambiguous cases, where several domains have similar heuristic rankings. Secondly, a classification layer significantly boosts up the accuracy since it distinguishes official websites, informative ones, and doubtful sources.

7. Conclusion

The present paper presented OWIP as a formal problem of web navigation by experts in authority and suggested a hybrid neuro-symbolic model, to address it. A combination of symbolic filtering and weighted ranking with neural semantic re-ranking provides the system with strong performance in noisy search environments. Future research will scale up the benchmark size and investigate multilingual and adversarial phishing cases.

8. References

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